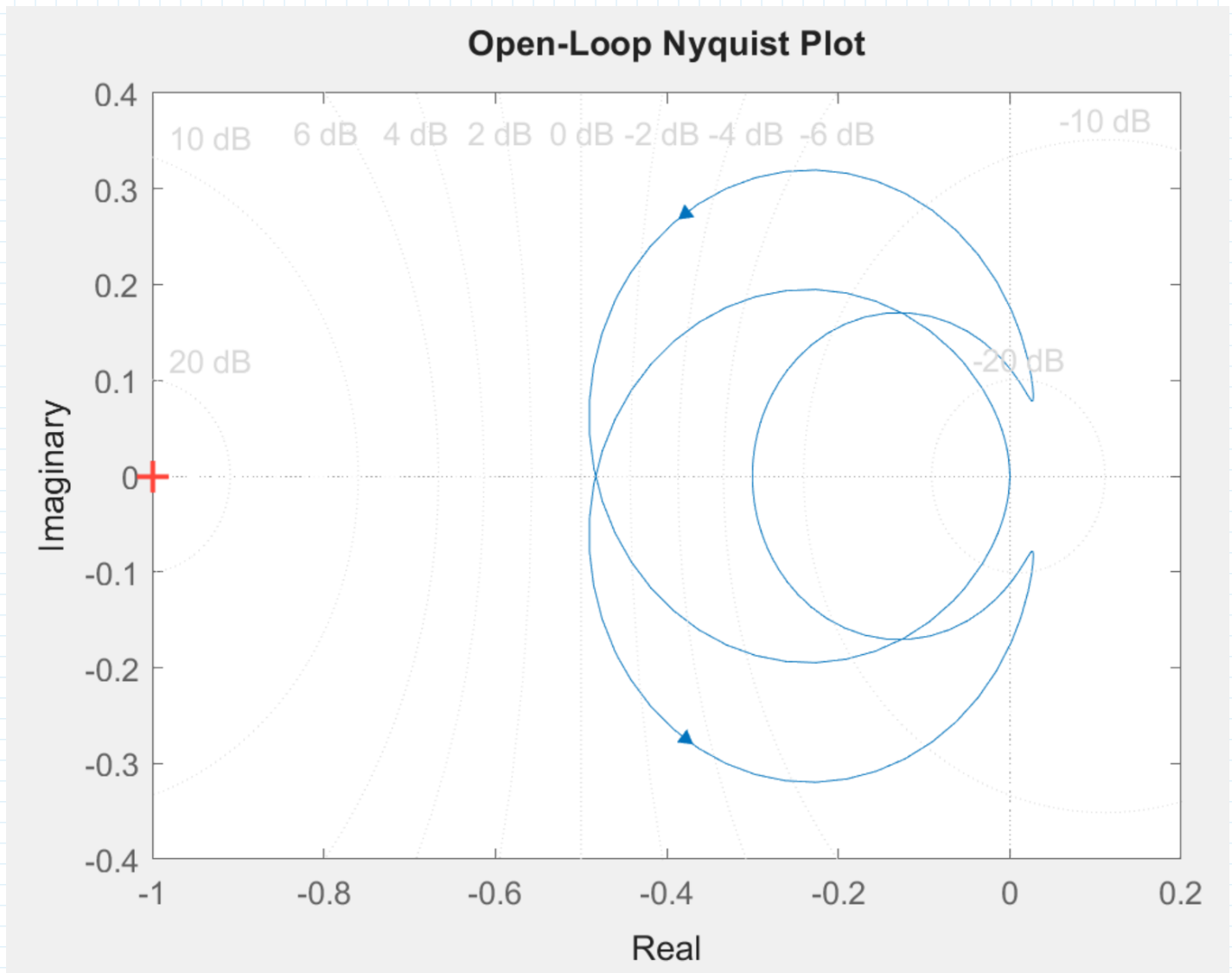


## Recap

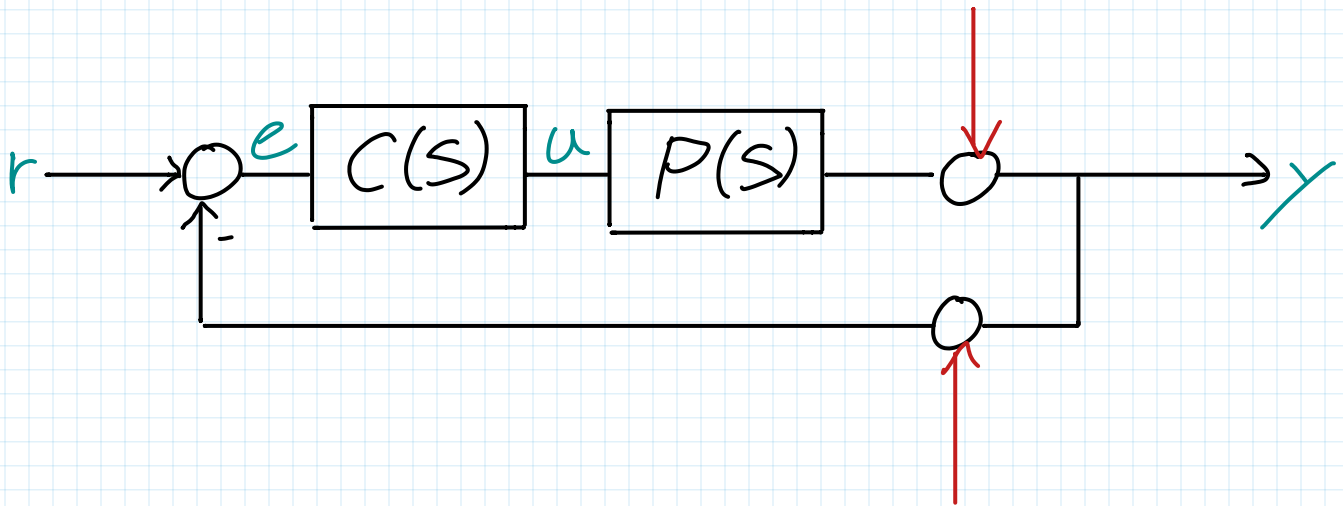
- Last week we introduced the Nyquist plot. With it, we were able to evaluate -loop stability, only by looking at the -loop tf.



- We saw that we not only want to be stable, but also !

$\Rightarrow$  Stable even in the presence of uncertainties and unwanted inputs

Let's take a closer look at how exactly our system can be affected:



The effect of  $n$  and  $d$  is seen in the tf's from:

$$\bullet \rightarrow : S(s) = \frac{1}{1 + C(s)P(s)}$$

$$\bullet \rightarrow : T(s) = \frac{C(s)P(s)}{1 + C(s)P(s)}$$

$\Rightarrow$  The effect of both  $n$  and  $d$  on  $e$  should be as   as possible.

- That means, both  $S(s)$  and  $T(s)$  should be as small as possible.

Theres one small issue...

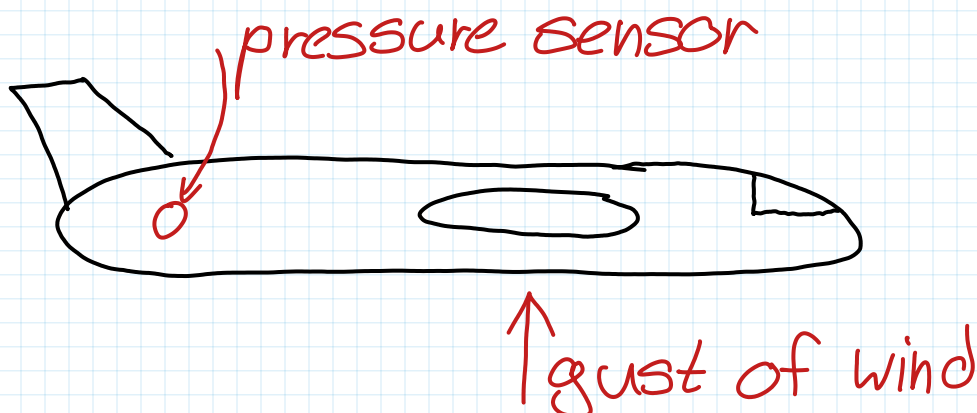
$$S(s) + T(s) = \frac{1}{1 + C(s)P(s)} + \frac{C(s)P(s)}{1 + C(s)P(s)} =$$

$\Rightarrow$  They always add up to one!

- This means that we cant cancel both noise and disturbances...

Thankfully they have a property we can make use of:

- Noise usually has a   frequency
- Disturbances usually have a   frequency.



• So the workaround is to require

1)  $\frac{1}{S(s)}$  for  $\frac{1}{S(s)}$   
 $\Rightarrow$  Disturbance rejection

2)  $\frac{1}{S(s)}$  for  $\frac{1}{S(s)}$   
 $\Rightarrow$  Noise rejection

How can we guarantee these constraints?

1)  $S(s) = \frac{1}{1 + G(s)}$   $\Rightarrow S(s)$  small, at small  $\omega$ .

Now we enforce the following inequality, using a  $\frac{1}{S(s)}$  function  $W_1(s)$

For some  $W_1(s)$ , that is large at low  $\omega$ .  
 The equation can be approximated as

$\Rightarrow$

$\approx$

(for small  $\omega$ )

$$2) T(s) =$$

$\Rightarrow T(s)$  small at high  $\omega$

Again using a constraint function:

For some  $W_2(s)$  that is large at high  $\omega$

• Again we rewrite and approximate:

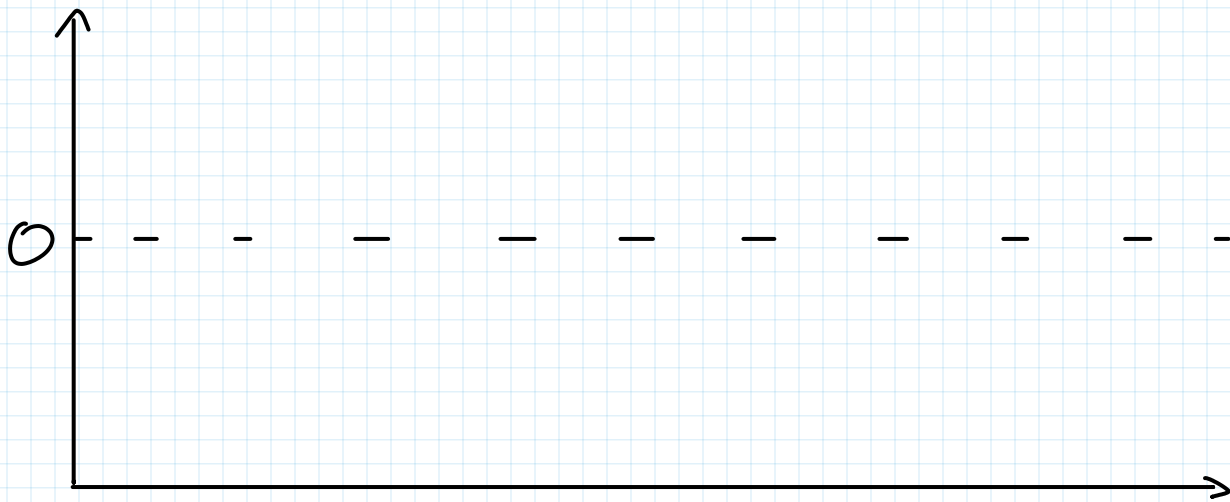
for high  $\omega$   
 $\approx$

$$\Rightarrow |L(s)| < |W_2(s)|^{-1}$$

What exactly are these  $W(s)$ ??

Their impact can be observed as

"                    " in the Bode plot as such:



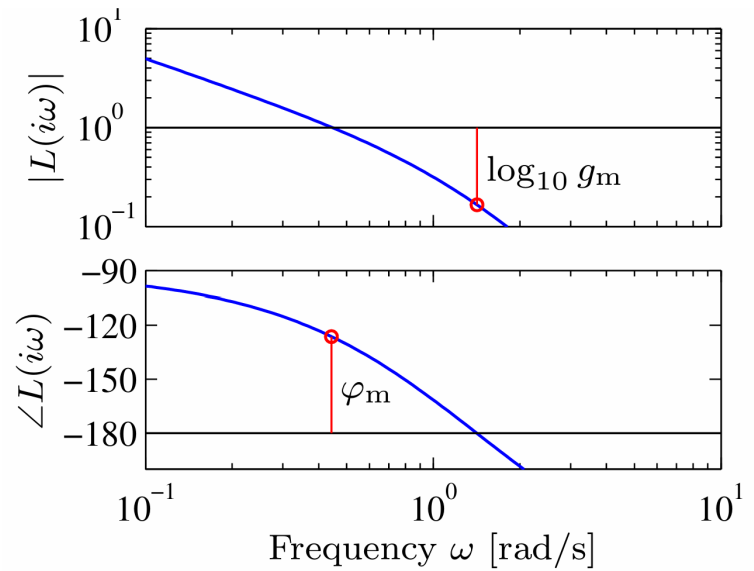
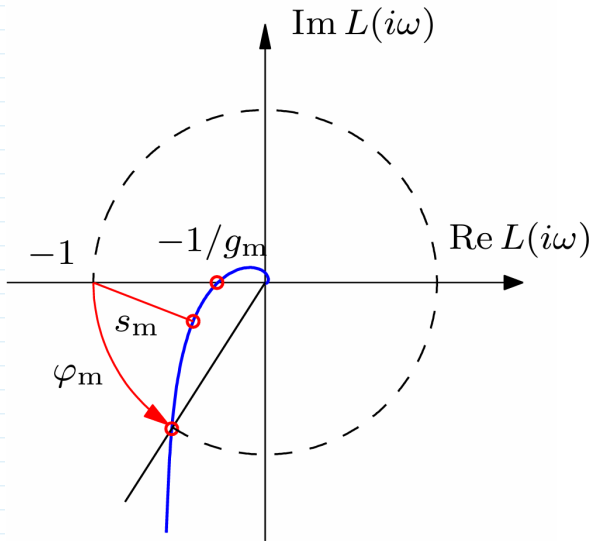
• The  $W$ 's are like obstacles for the open-loop tf, that *we can design*.

3) There's one more important condition for the closed loop:

$$T(s) = \frac{L(s)}{1 + L(s)}$$

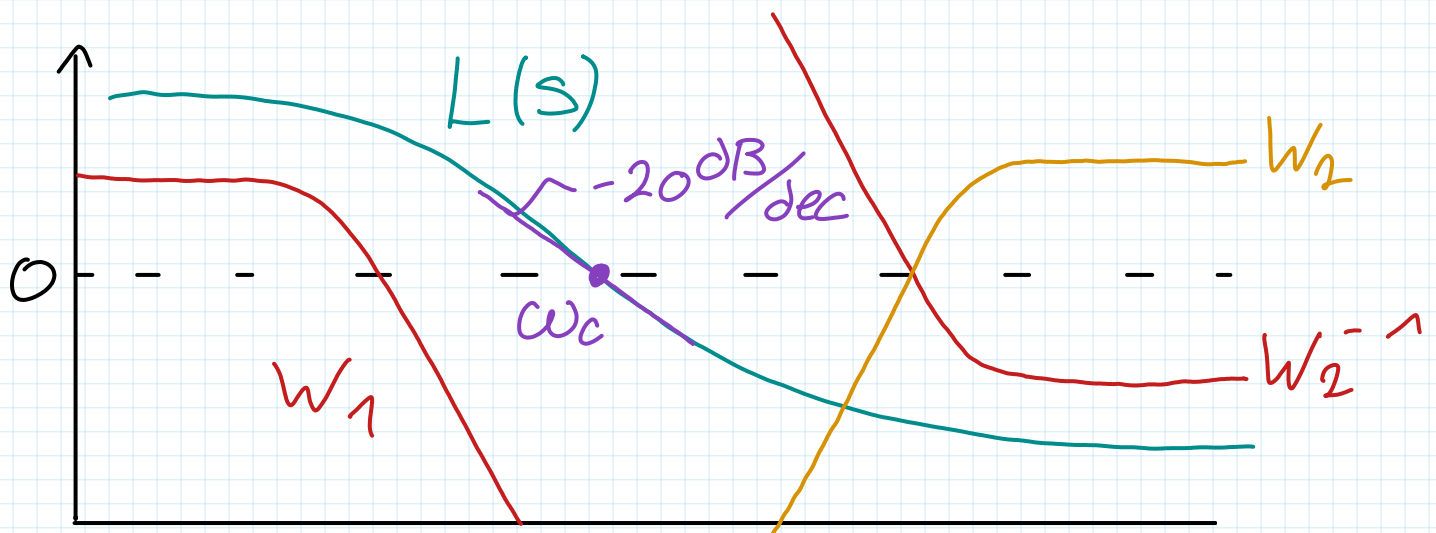
$\Rightarrow$  system becomes unstable for

$\Rightarrow$  We could also see this in the Bode-plot!



The slope of the magnitude plot is related to the phase!

- To ensure enough margin, the slope of the magnitude plot at the crossover frequency, should be at least -20 dB/dec or flatter.



4) The condition at the crossover frequency  $\omega_c$  is now established, but how do we even choose  $\omega_c$  itself?

$\Rightarrow$  We must take a little detour..

• Maximum frequency at which closed-loop can track references with a maximal attenuation of

$\Rightarrow \omega_b \uparrow$ : System reacts easily to fast signals

$\Rightarrow \omega_b \downarrow$ : System struggles to follow even slow signals

Choose  $\omega_b$ , s.t. that maximum signal speed you need to follow is still included.

From Bandwidth  $\omega_b$  to crossover  $\omega_c$

•  $\omega_b$  mathematically:

$$\omega \text{ s.t. } |T(j\omega)| = \frac{|L(j\omega)|}{|1 + L(j\omega)|} =$$

which is true for  $L(j\omega) =$

$$\begin{aligned} L(j\omega_b) &\Rightarrow |L(j\omega_b)| = \\ &\angle L(j\omega_b) = \end{aligned}$$

$\Rightarrow$  With a phase margin of ,

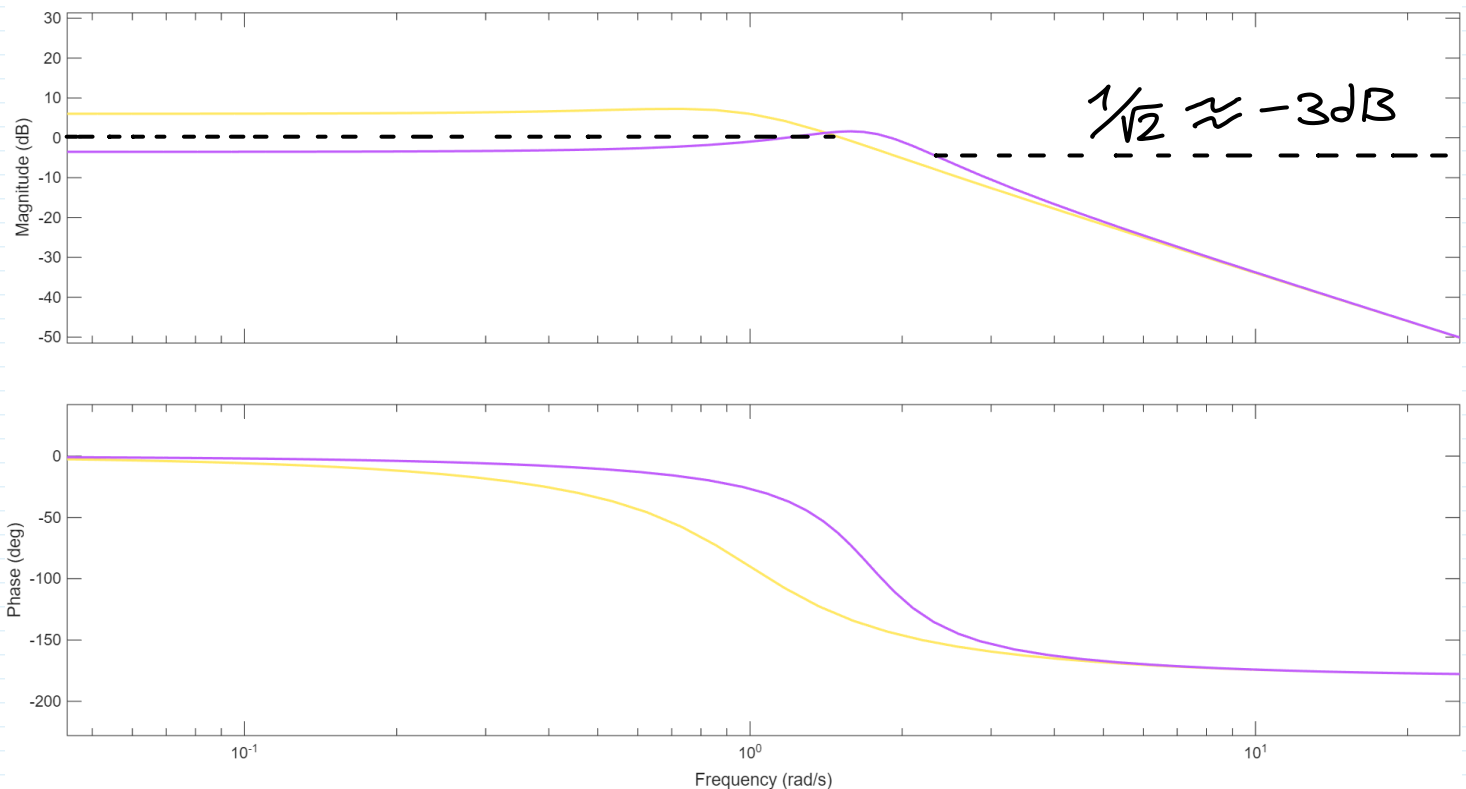
- Since we anyways desire  $\varphi_m \approx 80^\circ$ , we choose the open-loop  $P(s)$  such that the closed-loop requirement for  $\varphi_m$  is met.

### Example

Pendulum  $P(s) = \frac{2}{s^2 + s + 1}$  (linearized at  $(0,0)$ )

and closed-loop  $T(s) = \frac{P(s)}{1 + P(s)}$  (unit feedback)

Bode Diagram



$$\varphi_m =$$

$$=$$

$\Rightarrow$  Hence  $\omega_c \approx \omega_b$  and not  $\omega_c = \omega_b$

We have successfully formulated our closed-loop conditions solely on the open-loop  $L(s)$ .

So if we have a plant  $P(s)$  given, how do we now ensure the right frequency dependent behaviour of  $L(s)$ ?

- $L(s) = \text{_____}$  isn't going to work anymore...  
 $\Rightarrow$  We will need a  $\text{_____}$ !

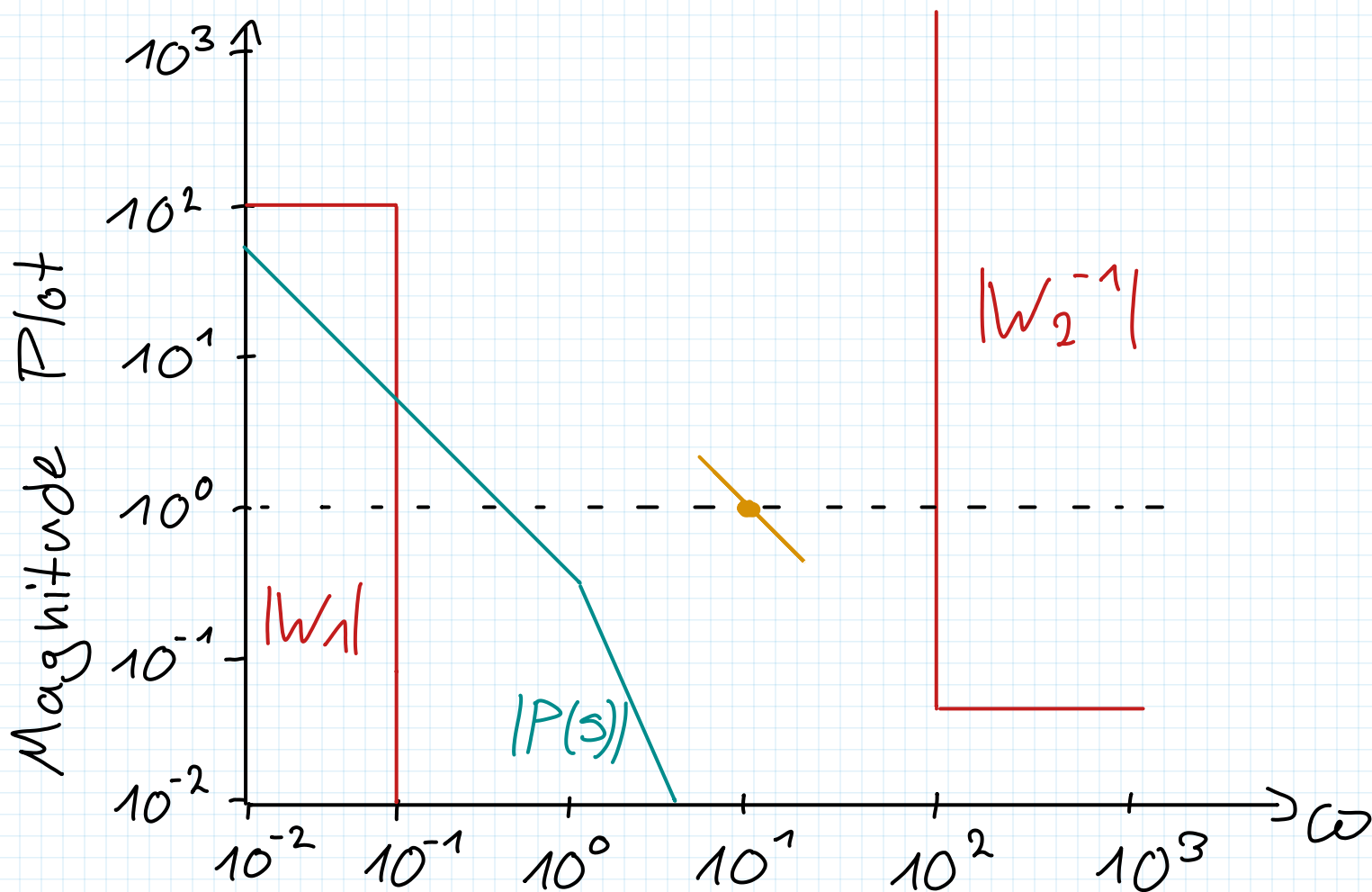
$$L(s) =$$

$\Rightarrow$  Choose the controller s.t.  $L(s)$  does what we want!

Ex: Given  $P(s) = 0,01 \cdot \frac{(s+10)^2}{s(s^2+2s+2)}$

Find a  $C(s)$ , s.t. the following conditions are respected:

- $W_1 = 100\text{dB}$  for  $\omega < 0,1 \text{ rad/s}$
- $W_2^{-1} = 0,05\text{dB}$  for  $\omega > 100 \text{ rad/s}$
- $\omega_c = 10 \text{ rad/s}$
- $45^\circ$  Phase margin at  $\omega_c$



- A good open-loop suggestion would be:

$$L_{des} =$$

- Now we reverse engineer:

$$C(s) =$$

What we just did was a "\_\_\_\_\_"  
approach...

- $C(s)$  may be unnecessarily \_\_\_\_\_
- Only works for open-loop stable, minimum-phase systems!
- A more sophisticated approach is called \_\_\_\_\_

$\Rightarrow$  Design a controller with the following elements, s.t.  $L(s)$  passes the obstacle course:

- Gain:
- Integrator:
- \_\_\_\_\_:
- \_\_\_\_\_:

$\Rightarrow$  These can be combined however we want. But is the effect of Lead/Lag-elements?

## Lead-Lag-Elements

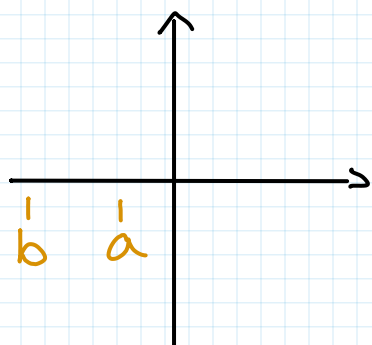
- In essence, Lead-Lag elements are a combination of a   and a  .

We use them to manipulate a frequency response

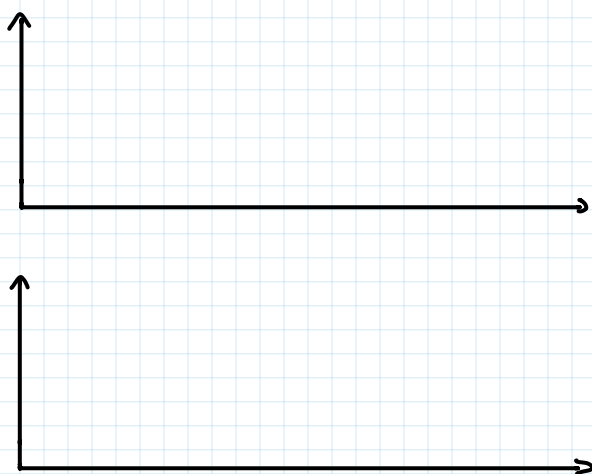
### Lead

$$C_{\text{lead}} = \frac{s/a + 1}{s/b + 1}, \quad 0 < a < b$$

• Roots:



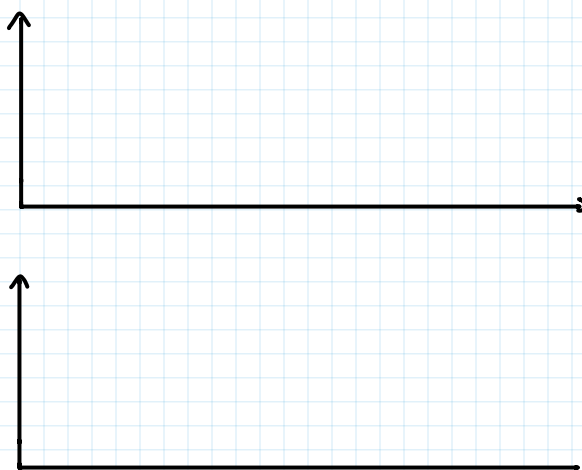
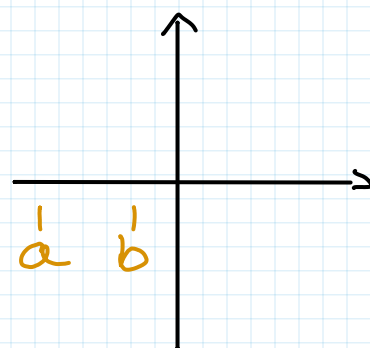
• Bode:



### Lag

$$C_{\text{lag}} = \frac{s/a + 1}{s/b + 1}, \quad 0 < b < a$$

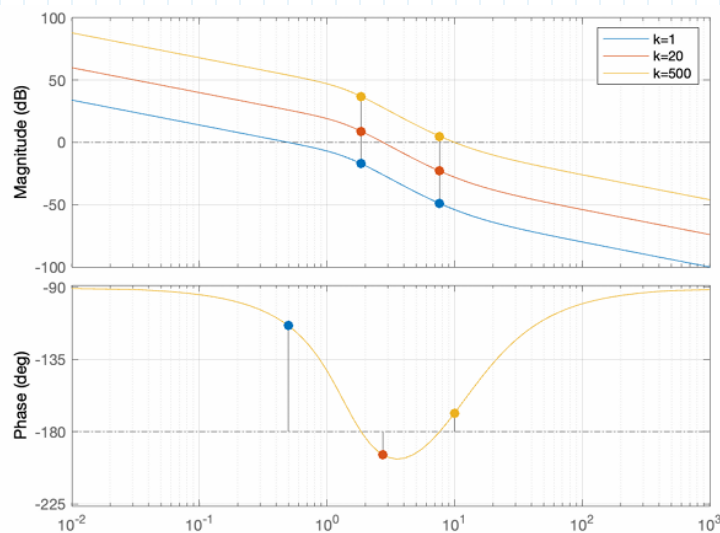
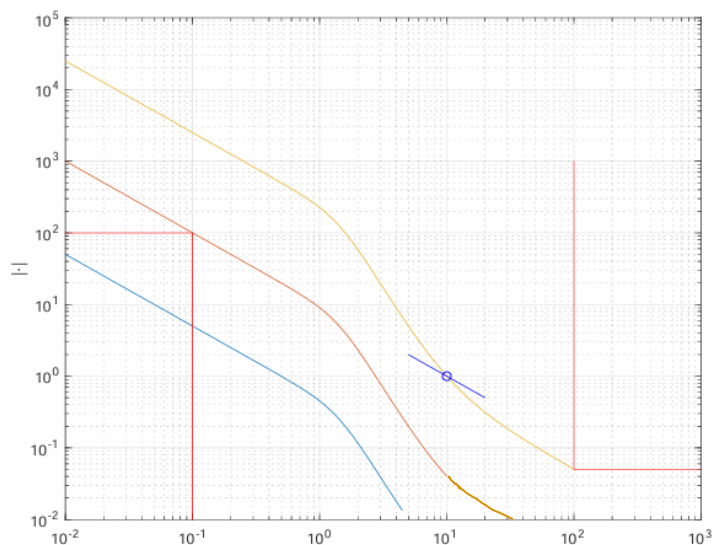
• Roots:



- magnitude by:   
 at high  $\omega$
- phase around  $\omega =$  , with max. increase:   
  $\varphi_{\max} =$
- magnitude by:   
 at high  $\omega$
- phase around:  $\omega =$  , with min. decrease:   
  $\varphi_{\min} =$

$\Rightarrow$  We can now build a controller brick by brick

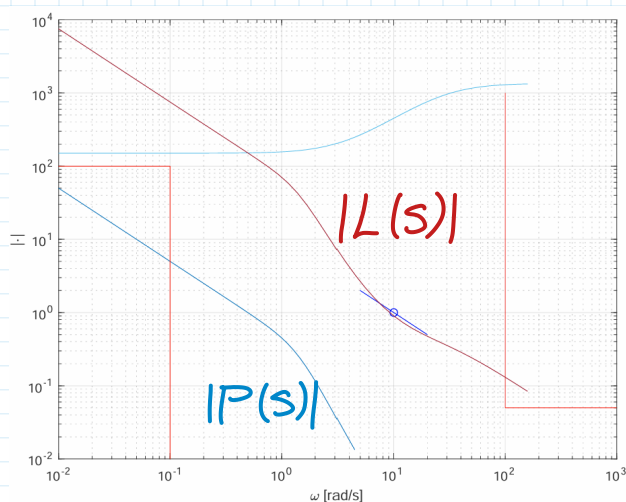
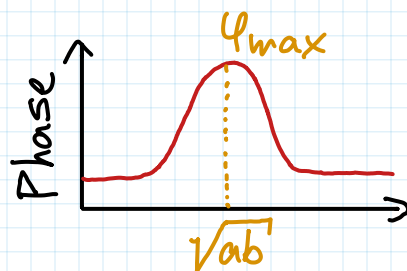
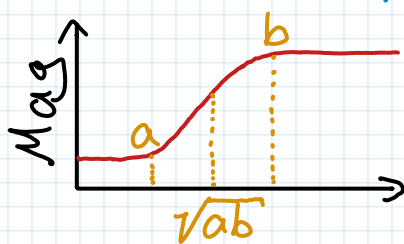
- First, look at the effect of  $k$ .



- $k=1$ : No rejection
- $k=20$ : point  $\Rightarrow$  unstable
- $k=500$ : Okay but tiny !

Let's settle for  $k=150$ . How do we address the missing phase margin and insufficient slope at the crossover frequency?

=> Add a  element!



• After rescaling  $k$ , we get:

$$C(s) = \underbrace{150}_K$$

How did we choose  $a, b$ ?

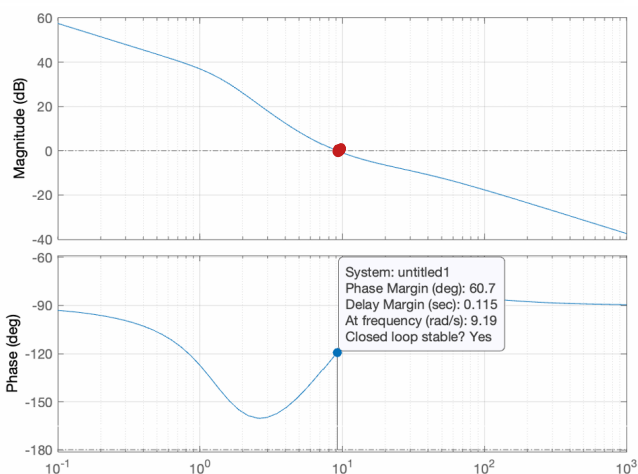
• Pick  at  $\omega_c$

=>

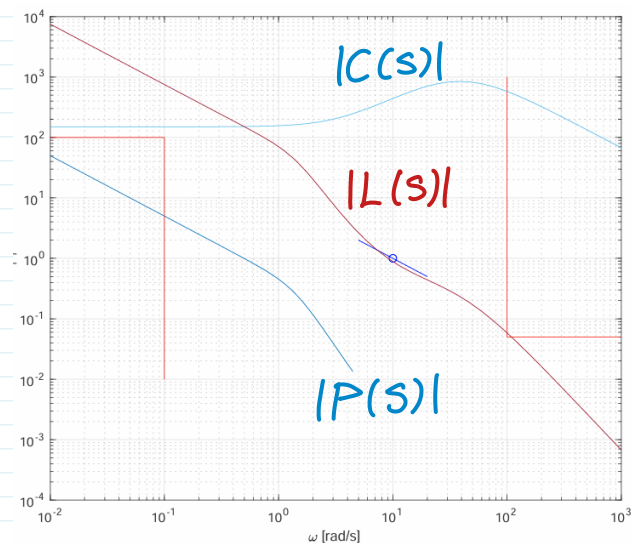
• Pick  by how much  $\phi_{\max}$  should be:

$$\phi_{\max} = 2 \arctan\left(\sqrt{\frac{b}{a}}\right) - 90^\circ$$

One problem left... We also need to   ... What element could we use for this?

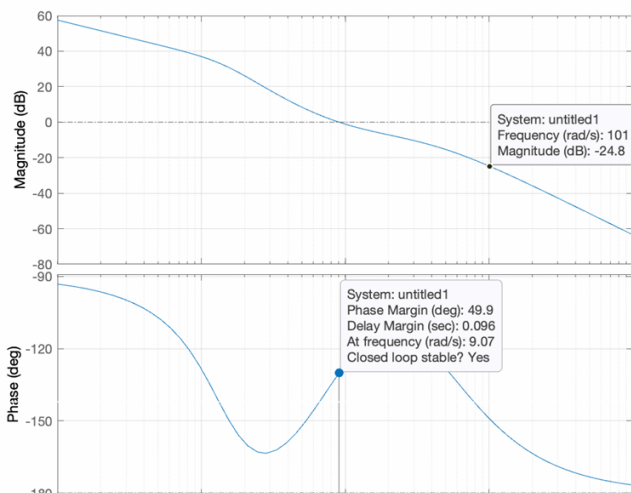


=> Add a   element!



$$C(s) = \underbrace{150}_k \cdot \underbrace{\frac{s/3.3 + 1}{s/30 + 1}}_{\text{lead}} \cdot \underbrace{\frac{s/1000 + 1}{s/50 + 1}}_{\text{lag}}$$

=> Now  $L(s)$  fulfills all demands!



Important note: All of the above works only for   stable systems...

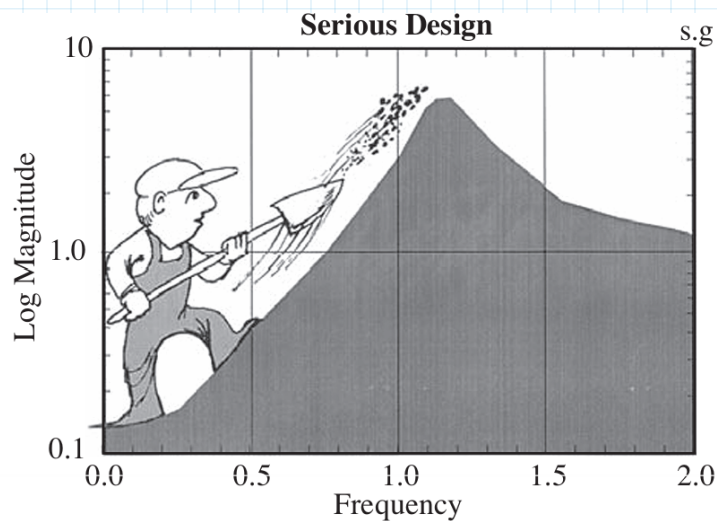
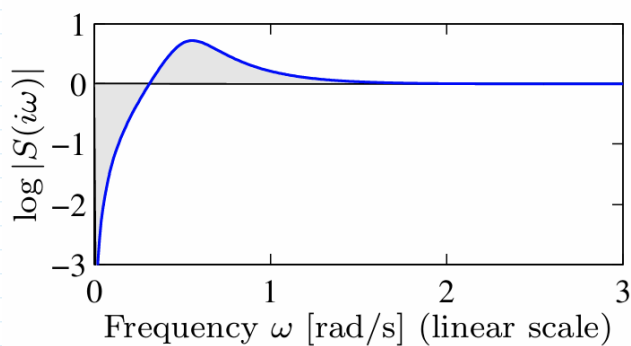
It seems as though we can just do what we want with our open-loop right?

## Bode's Integral

Let  $S(s)$  be the sensitivity function of a stable closed-loop. Then:

where  $p_k$  are the unstable poles of  $L(s)$ .

What does this mean?



Area underneath  $S(s)$  remains the same!

Let's look at an exam question:

**Problem:** Consider a lead compensator

$$C_{\text{lead}}(s) = \frac{s/a + 1}{s/b + 1}, \quad 0 < a < b, \quad (1)$$

used in a standard feedback architecture to control a given plant  $P$ .

**Q37 (0.5 Points)** Mark the correct answer for each statement.

| Statement                                                                                     | True | False |
|-----------------------------------------------------------------------------------------------|------|-------|
| A lead compensator is typically used to reduce the phase margin.                              |      |       |
| A lead compensator can increase the sensitivity to high-frequency noise.                      |      |       |
| The slope of the magnitude at frequencies between $a$ and $b$ is approximately +20 dB/decade. |      |       |

**Q38 (1.5 Points)** A lead compensator as given in Equation (1) is to be used in order to increase the phase of the open-loop system  $P(s) \cdot C_{\text{lead}}(s)$  by at most  $30^\circ$  at a frequency of 10 rad/s.

Which choice of the parameters  $a$  and  $b$  satisfies the aforementioned requirement? Mark the correct answer.

- ☐  $a = \frac{10}{\sqrt{3}}, b = 10\sqrt{3}$
- ☐  $a = 1\sqrt{3}, b = 100\sqrt{3}$
- ☐  $a = 10\sqrt{3}, b = \frac{10}{\sqrt{3}}$
- ☐  $a = 100\sqrt{3}, b = 1\sqrt{3}$

Q1)  $W_1$  &  $W_2$  make sure, that the closed loop is stable

☐ TRUE    ☐ FALSE

Q2) In general, the loop-shaping method produces less complex solutions for  $C(s)$  than the brute force approach

☐ TRUE    ☐ FALSE

Q3)  $C(s)$  always follows the "conservation of dirt" law (Bode's Integral)

☐ TRUE    ☐ FALSE